MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: PPL Roll No.: 03 Exp. No.: 02
Name of the Experiment:

Marks	Teacher's Signature with date
10	

Performed on:
Submitted on:

Experiment 2

Q.1 What is Inheritance in Python?

→ Inheritance is an object-oriented programming feature where one class acquires the properties and methods of another class. It allows: code reuse, logical hierarchy and extensibility.

Q.2 How does method overriding work?

→ A method overriding happens when a child class defines a method with the same name as a method in the parent class. The child's method replaces the parent's version.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Q.3 Why use error handling in this program?

→ Error handling is used to:

- Prevent the program from crashing
- Handle invalid user input.
- Maintain data integrity.
- Provide meaningful error messages.

Without it:

- Invalid input would cause runtime errors.
- The program would terminate unexpectedly.

Q.4. Can we add more attributes to the student class?

→ Yes we can easily add more attributes such as:

- Email
- course
- CGPA
- Address, etc.

Q.5 How can we improve this program?

→ Improvements include:

- Adding multiple student storage using a list or dictionary.
- Adding update and delete functionality.
- Implementing file handling

- Adding custom exception classes.
- Implementing multiple inheritance if needed.

Experiment 2- Create a Rectangle Area Calculator using a Rectangle class that accepts length and width and calculates area. Implement error handling to ensure dimensions are numeric and greater than zero, otherwise raise and handle appropriate exceptions.

Code-

```
class Rectangle:
    def __init__(self, length, width):
        if not isinstance(length, (int, float)) or not isinstance(width, (int, float)):
            raise TypeError("Length and width must be numeric values.")

        if length <= 0 or width <= 0:
            raise ValueError("Length and width must be greater than zero.")

        self.length = length
        self.width = width

    def calculate_area(self):
        return self.length * self.width


try:
    length = float(input("Enter length: "))
    width = float(input("Enter width: "))

    rect = Rectangle(length, width)
    area = rect.calculate_area()
    print(f"Area of rectangle: {area}")

except ValueError as ve:
    print("Value Error:", ve)

except TypeError as te:
    print("Type Error:", te)

except Exception as e:
    print("Unexpected Error:", e)
```

Output-

```
PS C:\Users\Microsoft\OneDrive\Desktop\abdulfile> & "C:/Program Files/Python314/python.exe" c:/Users/Micr
osoft/OneDrive/Desktop/abdulfile/assignment.py
Enter length: 5
Enter width: 10
Area of rectangle: 50.0
PS C:\Users\Microsoft\OneDrive\Desktop\abdulfile>
```